

MID SEMESTER – I EXAMINATION : 2023
BIT, SINDRI (B.TECH/ 1st SEMESTER)

Jharkhand University of Technology
(For Electrical, ECE, CSE & IT Branches only)

Full Marks : 20

Time : 1 ½ Hrs

Answer any FIVE out of the SEVEN questions as provided hereinunder. Try to write clearly in your own language as far as possible. Each question carries equal weightage of Marks.

- ✓ 1. What is the significance of introducing 'Indian Knowledge System' as a subject of study for Engineering Students under NEP 2020? CO1, CO2
2. Why do you think that India inspite of having Rich Ancestral Knowledge System and scientific legacy does not find much mention in Global Knowledge Base and Education System? CO3
3. What are the different Schools of Art that had flourished in Ancient India? Explain in details the period, main features of these Schools of Art and the places of their origins? CO1, CO3
4. Write short notes on any **two** of the following : CO1, CO2, CO4
 - (a) Patanjali
 - (b) Yoga in Ancient India
 - (c) Pottery and Utensils in Ancient India
5. Discuss the level of knowledge, expertise in the field of Astronomy and Mathematics in Ancient India. Do you think Indian Ancestral Astronomical and Mathematical knowledge/ work have contributed anything to the world of Science? CO1, CO2, CO6, CO7
- ✓ 6. How do you see Ayurveda as an alternate system of medicine? Elucidate in details the rich legacy of knowledge that India inherited in the field of Ayurveda from our ancestors in the form of seers, sages and saints. CO1, CO2
7. Critically examine the rich ocean of Ancestral Indian Knowledge System vis-a-vis its relevance to present day science and technology while focussing on its usefulness to mankind. CO1, CO6, CO7

B. Tech. 1st Semester (Mid Semester-I) Examination-2023

BIT Sindri, Dhanbad

Subject: Mathematics-I (All Branches)

Full Marks : 4x5=20

Time : 1 hr 30Mins

Subject Code : BSC 01

Answer any five questions.

1. (a) If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ then $A^2 - 4A + 3I$ is equal to [CO 1]

(i) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(iii) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(iv) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

- (b) The vectors $u=(6,2,3,4)$, $v=(0,5,-3,1)$, $w=(0,0,7,-2)$ are [CO 1]
 (i) Linearly dependent
 (ii) Linearly Independent
 (iii) Insufficient data
 (iv) None of these

- (c) The Eigen values of the matrix $A = \begin{bmatrix} 3 & 0 & 0 \\ 8 & 4 & 0 \\ 6 & 2 & 5 \end{bmatrix}$ are [CO 1]

(i) 3,4,5

(ii) 1,1,1

(iii) 2,3,6

(iv) None

- (d) The number of asymptotes of the curve $\frac{x^2}{4} + \frac{y^2}{9} = 1$ are [CO 2]

(i) 1

(ii) 2

(iii) 4

(iv) None

2. Convert the matrix $A = \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$ into normal form and hence find its rank. [CO 1]

3. Determine the characteristic roots and characteristics vectors of the $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$. [CO 1]

4. Investigate for what values of a and b, the equations $x+2y+3z=6$; $x+3y+5z=9$, $2x+5y+az=b$ have (a) no solution (b) a unique solution (c) Infinite solution. [CO 1]

5. Find the dimension of the subspace of R_4 spanned by the set $\{(1 \ 0 \ 0 \ 0), (0 \ 1 \ 0 \ 0), (1 \ 2 \ 0 \ 1), (0 \ 0 \ 0 \ 1)\}$. Hence find its basis. [CO 1]

6. Expand $\sin x$ in powers of $x - \frac{\pi}{2}$. Hence evaluate $\sin 91^\circ$ correct to four decimal places. [CO 2]

7. Find the asymptotes of the curve $y^3 - x^2y - xy^2 + x^3 - y^2 + x^2 = 0$. [CO 2]



DEPARTMENT OF CSE & IT
BIT SINDRI DHANBAD

First Mid-Semester Examination-2023

Subject – Programming for Problem Solving
Semester – 1st (CSE/IT/EE/ECE)

Time -1.5 Hour

Subject Code – ESPP1

Read all the questions carefully, do as directed. Attempt any 5 questions.

Write a program which takes two integer values from user, a number and a digit between 0 to 9 and find whether the digit is present in the number or not.

[CO1] [4 Marks]

Write a program to find the difference of the sum of all even numbers and the sum of all odd numbers present in an array.

[CO1, CO3] [4 Marks]

Write a program to print the below pattern where the number of lines to print will be taken from user.

```
1
123
12345
1234567
123456789
```

[CO1, CO3] [4 Marks]

Draw flowchart and write algorithm to compute the factorial of a number.

[CO2] [4 Marks]

Write a program to find the second smallest number in an array.

[CO1, CO3] [4 Marks]

What is operator? Write any five different types of operators in C with their precedence and associativity.

[CO1] [4 Marks]

Convert the below decimal numbers into their equivalent number of the given base.

- a. $(510)_{10}$ = $(\quad)_{16}$
b. $(-23)_{10}$ = $(\quad)_2$
c. $(49)_{10}$ = $(\quad)_2$
d. $(214)_{10}$ = $(\quad)_8$

[CO1] [4 Marks]

*****END*****

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HSM - 01 : Indian Knowledge System

MID SEMESTER - I EXAMINATION : 2023
BIT, SINDRI (B.TECH/ 1st SEMESTER)
Jharkhand University of Technology
(For Electrical, ECE, CSE & IT Branches only)

Full Marks : 20

Time : 1 ½ Hrs

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B. I. T. Sindri
P.O. Sindri Institute
Dhanbad- 828123

First mid semester Examination -2023.

Branch- Electrical Engg., E.C.E., C.S.E. and Information Technology.
Subject- Physics.

Full Marks - 20.

Answer any five questions. All question carry equal marks.

Time- 1½ Hours.

1×4=4.

1. Answer all questions.

(I) The wave function for the motion of the particle in a one- dimensional potential box of lengths is given by $\psi_n = A \sin\left(\frac{n\pi x}{a}\right)$. Where A is the normalization constant. The value of A is

- (a) $\frac{1}{a}$ (b) $\sqrt{\frac{2}{a}}$ (c) $\sqrt{\frac{a}{2}}$ (d) none of these.

Where a is the length of the box.

[CO1]

(II) In quantum mechanics a classical dynamical quantity is replaced by the corresponding quantum mechanical operator. Accordingly, the x component of momentum p_x is replaced by

- (a) $i\hbar \frac{\partial}{\partial x}$ (b) $-i\hbar \frac{\partial}{\partial x}$ (c) $i\hbar \frac{\partial}{\partial x}$ (d) none of these.

[CO1]

(III) The density of a simple cubic crystal is given by (a) $\frac{3M}{a^3 N_A}$ (b) $\frac{2M}{a^3 N_A}$ (c) $\frac{M}{a^3 N_A}$ (d) none of these.

[CO2]

Where terms have their usual meanings.

(IV) The atomic radius in a F.C.C. cell is

- (a) a (b) $\frac{a\sqrt{3}}{4}$ (c) $\frac{a}{2\sqrt{2}}$ (d) none of these.

[CO2]

Where terms have usual meaning.

OR

What do you mean by an operator? Find the operator corresponding to (a) Energy (b) linear momentum and (c) Kinetic energy.

1+3=4, [CO1]

2. (a) Derive the Schrodinger time dependent and time independent wave equation in one dimension. Write the Schrodinger time dependent and time independent wave equation in three dimensions.

(b) Explain the physical significance of the wave function.

2+1=4, [CO1]

3. Write the Schrodinger equation in one dimension for the particle in an infinite deep potential well

(rigid box) and solve it to obtain the Eigen values and Eigen function. Consider an electron in an infinite well of width

$0.5 \text{ \AA}$. Calculate energy of it in ground state.

1+2+1=4, [CO1]

4. What are Miller indices? How they are determined? Show in a diagram the (111) plane of a cubic lattice.

1+2+1=4, [CO2]

5. (a) Discuss the classification of solids into metals, semiconductors and insulators, on the basis of band theory.

(b) The unit cell of copper is a cube. The side of the cube is $3.6 \text{ \AA}$. If the unit cell in copper Solid are lined up side by side, how many unit cells will be there along 10mm length of the solid.

3+1=4, [CO2]

6. What are density of energy states in metals? Derive an expression for density of energy states of a metal.

1+3=4, [CO2]

7. Explain the followings (a) Expectation values (b) Eigen values and Eigen functions.

2+2=4, [CO1]

Subject kumar

Department of Electrical Engineering
BIT Sindri, Dhanbad-828123
B. Tech 1st Semester Mid Semester Examination 2023

Subject: Basics of Electrical Engineering (BEE)

Section: G, H & I

Full Marks= 4X5= 20

Subject Code: ESEE01

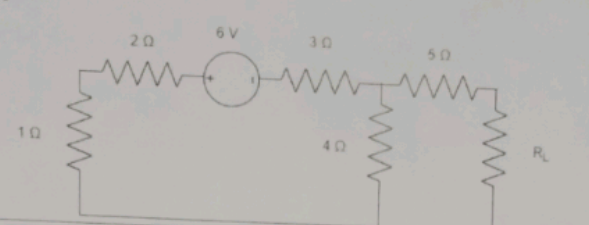
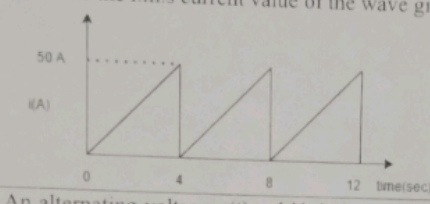
Branch: EE, ECE, CS&IT

Time: 1 Hour 30 Minutes

Instructions

- A. Answer **Any Five** questions
- B. Write in your own words.
- B. Each Question carries Equal marks.
- C. Assume missing data suitably.

Q.No.	Description	Marks	CO's
1 (a).	State and explain the Superposition theorem?	1	CO1
(b).	Determine the current in different resistors using superposition theorem in the circuit given below.	3	
2 (a).	State Thevenin's theorem?	1	CO1
(b).	Determine the V_{th} and R_{th} across R_L of the circuit given below	3	
3 (a).	Explain KVL and KCL?	2	CO1
(b).	Compute the equivalent resistance between terminals A & B using star-delta transformation	2	

4 (a).	State Maximum power transfer theorem?	1	
(b).	Determine the value of R_L which will absorb the maximum power from the circuit shown below. Also, calculate the value of maximum power?	3	CO1
			
5 (a).	Calculate the r.m.s current value of the wave given below?	2	CO1
			
(b).	An alternating voltage $v(t) = 141.4 \sin 314t$. Determine (i) frequency (ii) average value (iii) the time taken for the voltage to reach 100 V for the first time after passing through zero value.	2	
6 (a).	A 100 V, 60 W bulb to be operated from 220 V, AC supply. What resistance must be connected in series with the bulb to glow normally?	2	
(b).	A 230 V, 50 Hz voltage is supplied to an inductor of an inductance 0.2. Write the voltage and current equation. Also determine the maximum energy stored in the inductor?	2	CO1
7	Write short notes (any two) (i) Norton's theorem (ii) Star-delta transformation (iii) Average value of an alternating quantity (iv) Compute current and power of the single-phase a.c circuit containing inductance only (v) Compute current and power of the single-phase a.c circuit containing capacitance only	2x2	CO1
